

Problem

The step response of a parallel RLC circuit is $v = 10 + 20e^{-300t}(\cos 400t - 2 \sin 400t)$ V, $t \geq 0$ when the inductor is 25 mH. Find R and C .

Step-by-step solution

Step 1 of 4

Consider the expression for the voltage $v(t)$.

$$v(t) = 10 + 20e^{-300t}(\cos 400t - 2 \sin 400t) \text{ V}$$

Compare with $v(t) = v(\infty) + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t}$.

The values of α and ω_d are 300 Np/s and 400 rad/s.

Step 2 of 4

Calculate the value of the natural frequency ω_o .

$$\omega_o = \sqrt{\omega_d^2 + \alpha^2}$$

Substitute 400 rad/s for ω_d and -300 Np/s for α .

$$\begin{aligned}\omega_o &= \sqrt{200^2 + 300^2} \\ &= \sqrt{160000 + 90000} \\ &= \sqrt{250000} \\ &= 500 \text{ rad/s}\end{aligned}$$

Therefore, the expression for the natural frequency ω_o is 500 rad/s.

Step 3 of 4

Consider the expression for the natural frequency ω_o of the parallel connected RLC circuit.

$$\omega_o = \frac{1}{\sqrt{LC}}$$

Substitute 500 rad/s for ω_o and 25 mH for C .

$$\begin{aligned}500 &= \frac{1}{\sqrt{(25 \text{ m})C}} \\ 250000 &= \frac{1}{(25 \text{ m})C} \\ C &= \frac{1}{(25 \text{ m})(250000)} \\ C &= \frac{1}{6250} \\ C &= 160 \mu\text{f}\end{aligned}$$

Therefore, the value of the capacitance C is $160 \mu\text{f}$.

Step 4 of 4

Consider the expression for the neper frequency α .

$$\alpha = \frac{1}{2RC}$$

Substitute -300 Np/s for α and $160 \mu\text{f}$ for C .

$$300 = \frac{1}{2R(160 \mu)}$$

$$R = \frac{1}{2(300)(160 \mu)}$$

$$R = \frac{1}{0.096}$$

$$R = 10.4167 \Omega$$

Therefore, the value of the resistance R is 10.4167Ω .